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III. NETWORK TRAINING

In this study, the training is based on the open source neural network framework Darknet and the YOLOv3 network structure as a model. In order to speed up the training and have the high accuracy, the momentum constant is set to 0.9 and the decay coefficient is 0.0005, and In order to reduce the training time, the max batches will reduce to 2000. This study is trained by the Darknet53 network model to increase the accuracy and detect the smaller objects.

A. DATA SET

The data set used in this study is collected by ourselves, which is the most commonly used to the users, All particular object data set has 1069 images, there has 996 positive images and 73 negative images, the reason why we put the 73 negative images is that sometime will have the wrong detection.

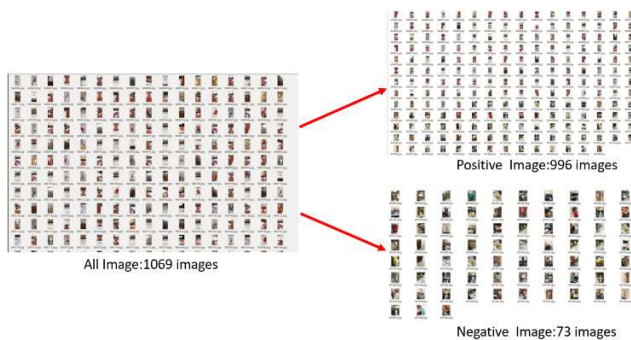


Fig. 2. The particular data set in training

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This study has trained for three section, and the software and hardware platform used in this study is shown in Table I. the dataset has included 2 classes, and the final dataset contains 1069 images, and it is composed of 60% for training ,In this experiment, it has been through 3 section, all section use the YOLOv3.cfg to train, the batch was set at 64 and the subdivision was set at 16. In order to reduce the training, the max batches was set at 2000, the first experiment dataset only have 200 images and it can't detect the particular object after training, the reason why it can't detect the object is that when we prepare the training data set, the training image has been rotated, so the origin image can't contrast to the training image, the result of detection is shown in Fig3 and the average loss is shown in Fig4.

Hardware Platform		Software Platform	
CPU	IntelCore i7-8700	Operating System	Ubuntu 18.04
GPU	RTX 2080ti	Deep Learning Framework	Darknet
RAM	16G	Programing Language	Python



Fig. 3. The first training result

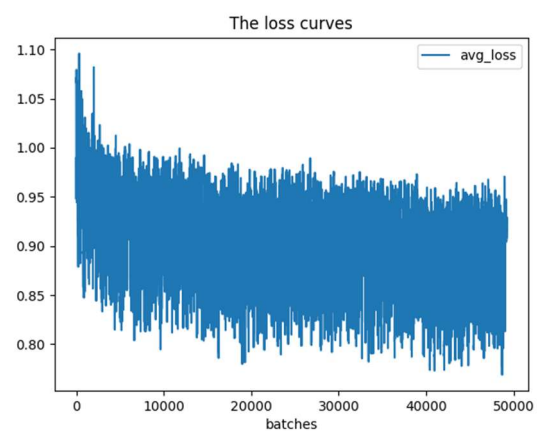


Fig. 4. The first training average loss

The first training has a worst detection result, it can't detect the particular object, so the problem has happened in the first training is been fixed in the second training, In order to let the detection has higher accuracy, the object dataset will increase to 550 images, and the second training has a better results.

In the second training, It is successful to detect the particular object, but the detection distance is too close and the accuracy is not enough.

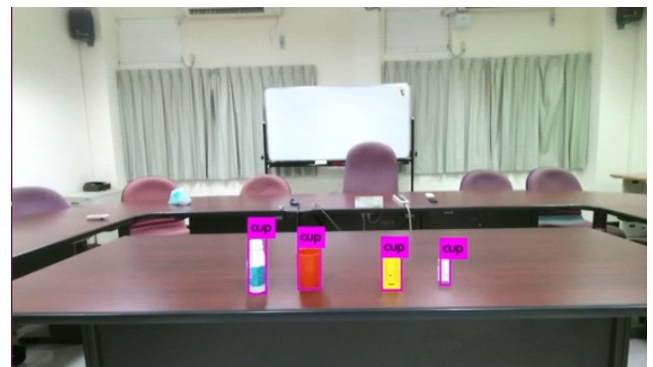


Fig. 5. The second training result

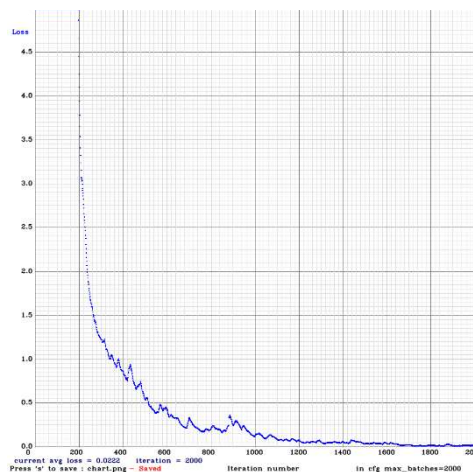


Fig. 6. The second training average loss

In order to improve those issue, the third training add more positive training images and the negative training images to increase the accuracy, the third training has good result to detect the far particular object, and the accuracy is enough for the service robots.



Fig. 7. The third training result

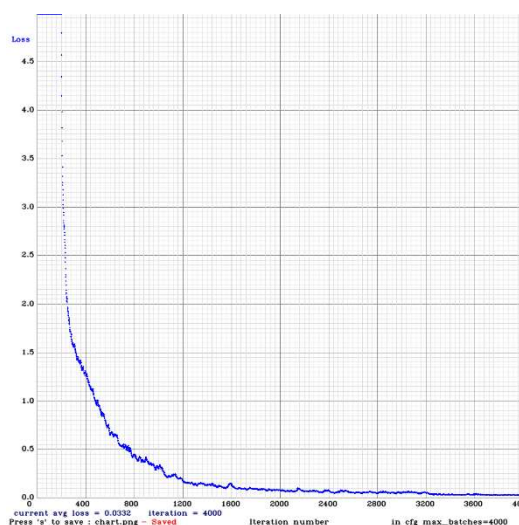


Fig. 8. The third training average loss

In this study, we use the weights of final training to detect the cup and bottle on the autonomous service robot, the hardware use on the service robot is NVIDIA Jetson AGX Xavier, the number of detection frames on AGX Xavier can reach 10 frames, it is enough for the service robot to take the particular object, the Fig.9 and Fig.10 is the picture of the experiment on the service robot.

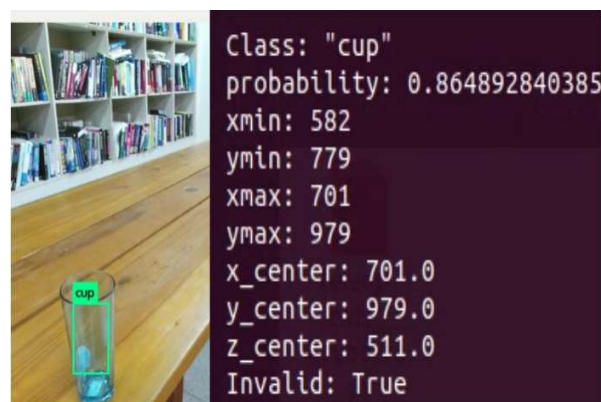


Fig. 9. The detection on the robot

The Fig.9 is the detection of YOLO on the robot, it figures out the class of the object, the probability, the middle of the image detection, and the x, y maximum.

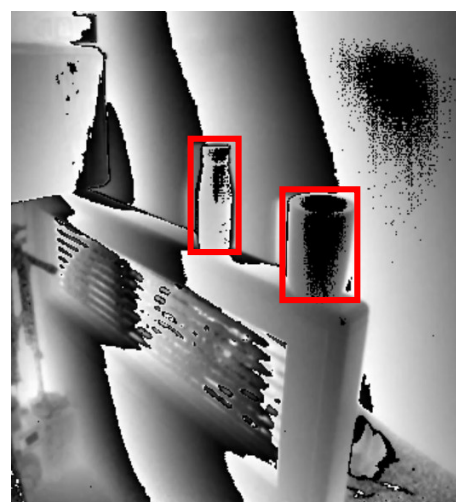


Fig. 10. The object detect by Kinect sensor

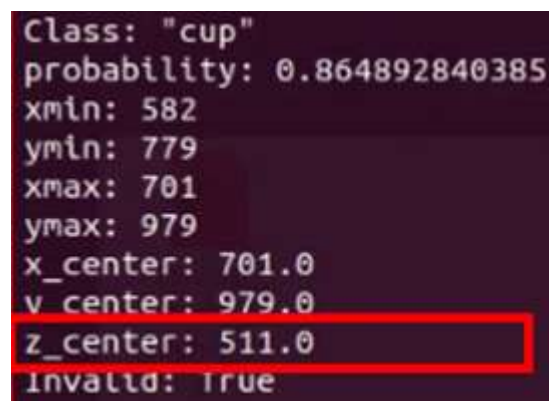


Fig. 11. The value of depth information

In order to let the service robot could exactly take the object, we use the Kinect sensor to get the deep information, and In the Fig11, the value of z center is the deep information which detect by the Kinect sensor, and we will translate the value to the real distance between the robot and the object. The service robot would use the x, y maximum and the deep information to take the object successfully.



Fig. 12. The autonomous service robot



Fig. 13. The autonomous service robot

V. CONCLUSION

This paper study the training of the YOLO object recognition, it has achieved good results in particular object detection. In this paper, the training has been through three section, and every section have some issues, and it has been solved at the end, we use the experiment result on the autonomous service to detect the particular object, and the robot could successfully take the object to the user.

ACKNOWLEDGMENT

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